



UAI301 Object Oriented Programming

Lab Assignment 1

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Sub-Group: 2X11

Subject Code: UAI301

Subject Name: Object Oriented Programming

Question 1: Printing Text Write a Python program to display the following information on the screen:

- Your Name

- Department

- College Name

```
In [30... print("Name: Amritanshu Mishra")
print("Department: Computer Science and Engineering")
print("College Name: Thapar Institute of Engineering and Technology")
```

Name: Amritanshu Mishra

Department: Computer Science and Engineering

College Name: Thapar Institute of Engineering and Technology

Question 2: Arithmetic Operations Write a Python program to display the results of the following arithmetic expressions:

- 15×6

- $45 - 18$

- $24 + 36$

- $81 \div 9$

```
In [31... print("15 x 6 =", 15 * 6)
print("45 - 18 =", 45 - 18)
print("24 + 36 =", 24 + 36)
print("81 ÷ 9 =", 81 / 9)
```

$15 \times 6 = 90$

$45 - 18 = 27$

$24 + 36 = 60$

$81 \div 9 = 9.0$

Question 3: Using Variables Write a Python program that stores the following details in variables and displays them.

- Student Name

- Branch
- Semester
- University Name

```
In [32... name = "Rahul Sharma"
branch = "CSE"
semester = 3
university = "ABC University"

print("Name:", name)
print("Branch:", branch)
print("Semester:", semester)
print("University:", university)
```

Name: Rahul Sharma
Branch: CSE
Semester: 3
University: ABC University

Question 4: Basic Mathematical Calculations Write a Python program to display the results of the following operations:

- $125 + 75$
- $200 - 85$
- 18×12
- $144 \div 12$

```
In [33... print("125 + 75 =", 125 + 75)
print("200 - 85 =", 200 - 85)
print("18 x 12 =", 18 * 12)
print("144 ÷ 12 =", 144 / 12)
```

$125 + 75 = 200$
 $200 - 85 = 115$
 $18 \times 12 = 216$
 $144 \div 12 = 12.0$

Question 5: Accepting User Input Write a Python program to accept the user's name from the keyboard and display it on the screen.

```
In [34... name = input("Enter your name: ")
print(name)
```

Amritanshu

Question 6: Greeting the User Write a Python program that accepts the user's name and displays a welcome message.

```
In [35... name = input("Enter your name: ")
print("Welcome", name)
```

Welcome Amritanshu

Question 7: Addition of Two Numbers Write a Python program to input two integers from the user and display their sum.

```
In [44... num1 = int(input("Enter first number: "))
num2 = int(input("Enter second number: "))

print("Sum =", num1 + num2)
```

Sum = 25

Question 8: Employee Information Write a Python program to accept the following details from the user:

- Employee Name
- Years of Experience
- Department

Display all the entered information in a neat format.

```
In [37... name = input("Employee Name: ")
experience = int(input("Years of Experience: "))
department = input("Department: ")

print("\nEmployee Details")
print("Employee Name:", name)
print("Experience:", experience, "years")
print("Department:", department)
```

Employee Details
Employee Name: Rahul Garg
Experience: 8 years
Department: Tech

Question 9 Write a Python program to display your:

- Name
- Father's Name
- Roll Number
- Branch
- Semester using variables.

```
In [38... name = "Arnav Garg"
father_name = "Ajay Garg"
roll_number = "102403001"
branch = "COE"
semester = 3

print("Name:", name)
print("Father's Name:", father_name)
print("Roll Number:", roll_number)
print("Branch:", branch)
print("Semester:", semester)
```

Name: Arnav Garg
Father's Name: Ajay Garg
Roll Number: 102403001
Branch: COE
Semester: 3

Question 10 Write a Python program to accept the following details from the user: Name, Roll Number, Branch and Semester. Display all the information using the print() function.

```
In [39... name = input("Enter Name: ")
roll = input("Enter Roll Number: ")
branch = input("Enter Branch: ")
semester = input("Enter Semester: ")

print("\nStudent Details")
print("Name:", name)
print("Roll Number:", roll)
print("Branch:", branch)
print("Semester:", semester)
```

Student Details
Name: Ajay Sinha
Roll Number: 10240031
Branch: COE
Semester: 3

Question 11 Write a Python program that accepts two numbers from the user and displays:

- Sum
- Difference
- Product
- Quotient

```
In [40... num1 = float(input("Enter first number: "))
num2 = float(input("Enter second number: "))

print("Sum =", num1 + num2)
print("Difference =", num1 - num2)
print("Product =", num1 * num2)
print("Quotient =", num1 / num2)
```

Sum = 19.0
Difference = 5.0
Product = 84.0
Quotient = 1.7142857142857142

Question 12 Write a Python program to accept your favorite programming language as input and display the message: My favorite programming language is Python. (Replace Python with the user's input.)

```
In [41... language = input("Enter your favorite programming language: ")
```

```
print("My favorite programming language is", language + ".")
```

My favorite programming language is C++.

Question 13 Write a Python program that accepts the student's name, branch, semester, and CGPA from the user and displays all the details in the following format: **** Student Details ****

Name : __

Branch : __

Semester : __

CGPA : __

```
In [42... name = input("Enter Name: ")
branch = input("Enter Branch: ")
semester = input("Enter Semester: ")
cgpa = float(input("Enter CGPA: "))

print("***** Student Details *****")
print("Name :", name)
print("Branch :", branch)
print("Semester :", semester)
print("CGPA :", cgpa)
print("*****")
```

```
***** Student Details *****
Name : Amritanshu
Branch : DSAI
Semester : 3
CGPA : 8.5
*****
```

Question 14 Write a Python program to accept three numbers from the user and display:

- Sum
- Average

```
In [43... num1 = float(input("Enter first number: "))
num2 = float(input("Enter second number: "))
num3 = float(input("Enter third number: "))

sum = num1 + num2 + num3
average = sum / 3

print("Sum =", sum)
print("Average =", average)
```

```
Sum = 79.0
Average = 26.333333333333332
```